

Rosas Behoundja

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Artificial Intelligence student passionate about using combinatorial optimization and machine learning to build practical tools that solve real-world challenges across Africa.

EDUCATION

IFRI, University of Abomey-Calavi, Bénin

Nov 2023 - Present

Bachelor of Science in Artificial Intelligence.

Expected completion: november 2026

EXPERIENCE

Applied Machine Learning Mentor

Bénin

Sèmè City

July 2026

- Mentored the top 20 participants of the National Olympiad of Artificial Intelligence (NOAI) 2026, providing guidance in Machine Learning and supporting them throughout their technical challenges.

Research Intern

Bénin

Computer Science and Applications Research Laboratory of IFRI (LRSIA) [🔗](#)

March 2026 – August 2026

- Designed a MILP model for the MPVRP variant focusing on changeover costs (MPVRP-CC); initiated Constraint Programming modeling of the problem using successor and sequence variables.
- Supervised the second-year AI undergraduate class project (Machine Learning course), contributing to the [ifri-mini-ml-lib](#) [🔗](#) pedagogical library (neural network, NLP, and anomaly detection modules).
- Contributed to the lab's internal projects and to technology monitoring.
- **Tools:** Gurobi, MaxiCP, MiniCP, Python, Java

AI Developer

Bénin

IFRI & Tekbot Robotics [🔗](#)

June 2025 – Jan 2026

- Developed and deployed an AI vision model and robotic controls for an automated waste-sorting system, allowing a robot arm to instantly recognize and sort trash in real time to improve urban recycling.
- **Tools:** CNN, YOLOv5, PyTorch, Jetson Nano, CUDA, ONNX, TensorRT

Machine Learning Engineer Intern

South Africa

eTihuku [🔗](#)

June 2025 – Aug 2025

- Built Review Scraper, an asynchronous API for collecting and aggregating customer reviews from Google Maps, HelloPeter, and X (Twitter), with background task processing and a scalable architecture.
- Worked on SentiMaster, an AI platform that collects customer feedback from social media and phone calls, automatically translates and analyzes hidden sentiments, emotions and topics, and helps companies predict customer churn.
- **Tools:** FastAPI, Playwright, PyTorch, Transformers, BERT models, Airflow, Redis

SELECTED PROJECTS

- **le-foncier-intelligent** [🔗](#): Developed a geospatial land-analysis pipeline combining multimodal OCR, geocoding, and satellite imagery to extract and assess information from topographic documents. Processed geospatial data using GeoPandas, Shapely and pyproj to support automated land-use analysis.
- **automated real estate mapping** [🔗](#): Designed of an ETL pipeline to link address data, cadastral information, companies, and executives, in order to automatically generate enriched datasets for prospecting and analysis.
- **MDVRP** [🔗](#): Designed and implemented optimization algorithms (ALNS, PSO, Simulated Annealing, Linear Programming) for multi-depot vehicle routing (MDVRP) under capacity and distance constraints.
- **Opti'plan**: Co-built an automated scheduling model that reduced the university's thesis defense planning time from 3 days to under 5 minutes using Constraint programming.
- **insurance-recommandation-system** [🔗](#): Developed an end-to-end machine learning pipeline for insurance prod-

uct recommendation, including data processing, model training and comparison (XGBoost, CatBoost, Deep Learning), with a Flask API and web interface supporting prediction and automated model retraining.

- **ifri-mini-ml-lib** [🔗](#): Co-built an open-source Python library that reimplements classic AI algorithms from scratch for educational purposes; published on PyPI and used by IFRI students.

ENGAGEMENTS

THE LAB & FRIARE Africa: Co-lead an AI awareness community in Bénin; participate in awareness initiatives on AI ethics, transparency, inclusion.

IFRI AI & IoT Club: Co-lead the AI & IoT student club; deliver introductory sessions on Machine Learning and Optimization Algorithms, coordinate technical activities, and co-supervise AI projects ([GitHub](#) [🔗](#)).

AWARDS

1st Prize – IndabaX Benin Republic Hackathon 2024

[Certificate](#) [🔗](#)

2nd Place – Tekbot Robotics Challenge 2025

[Link](#) [🔗](#)

TALK, POSTERS AND CONFERENCES

08/2026 – Deep Learning Indaba: [🔗](#) Presented a poster on A MILP Framework for Multi-Product Vehicle Routing Problem with Changeover Costs.

11/2025 – Benin Workshop on AI: [🔗](#) Introduced evolutionary algorithms concepts and their applications in optimization.

04/2026 – AMLD 2026: [🔗](#) Co-organized the first edition of the Applied Machine Learning Days in Benin Republic.

08/2025 – ACP Summer School: [🔗](#) Participated in the Association for Constraint Programming summer school, working on CP for sustainable development.

11/2024 – IndabaX Benin Republic: [🔗](#) Won the hackathon.

10/2024 – Future of AI Community: [🔗](#) Presented GPU Programming with CUDA for AI.

SKILLS

Programming & Data	Python, Java, R
ML/AI	Jax, PyTorch, Scikit-learn, Gurobi, Azure, ROS2
Tools	Git, Linux/Bash, Docker, FastAPI, GitHub Actions
Soft Skills	Teamwork, Leadership, Communication, Problem solving, Critical thinking.
Languages	French - (Native) · English - (<i>B2 Upper Intermediate</i>) 🔗

RESEARCH INTERESTS

Combinatorial optimization, industrial planning and scheduling problems, applied machine learning for social good, speech and language processing for low-resource languages.